

Regulatory Genomics

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Roadmap of STAT/BIOF/GSAT540 2026

data generation followed by *methodology*

- ③ Formulating biological questions
- ④ RNA-seq data
- ⑤ Experimental design
- ⑥ Single-cell genomics
- ⑦ Matrix factorization
- ⑩ GWAS
- ⑪ **Regulatory genomics** (Today)
- ⑫ Regression (for reg. genomics)
- ⑬ Biological networks
- ⑭ Clustering

Today's lecture

- 1 Technologies for measuring regulatory activities
- 2 DNA Methylation (credit: Keegan Korthauer)
- 3 Multiomic Data Integration with Regulatory Genomics

Chromatin Immunoprecipitation + Sequencing

Goal: Identify where proteins (e.g., TFs) bind to DNA genome-wide

Experimental Protocols:

- ① Cross-link proteins to DNA
- ② Fragment chromatin
- ③ Immunoprecipitate with antibody
- ④ Sequence DNA fragments
- ⑤ Map reads to genome
- ⑥ **Call peaks**

How ChIP-seq works

a DNA-binding protein ChIP-seq

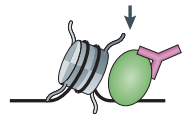


Crosslink proteins and DNA



Sample fragmentation

- Sonication
- Endonuclease (ChIP-exo)



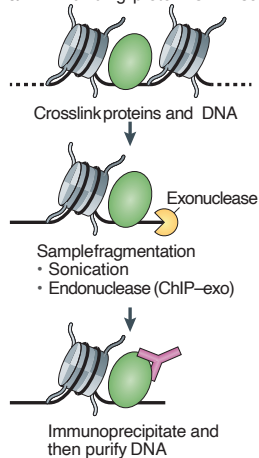
Immunoprecipitate and
then purify DNA

- Binding
- Sequencing

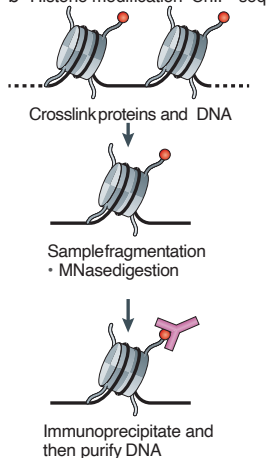
Furey (2012)

How ChIP-seq works

a DNA-binding protein ChIP-seq



b Histone modification ChIP-seq

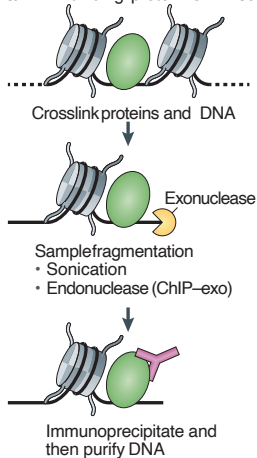


- Binding
- Sequencing

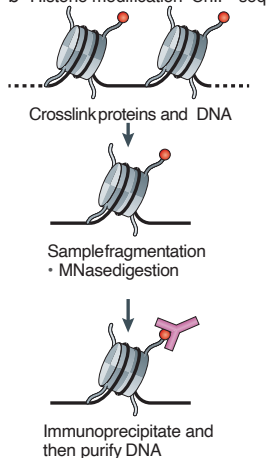
Furey (2012)

How ChIP-seq works

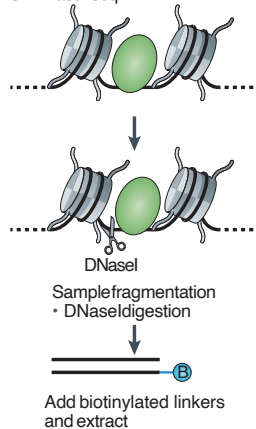
a DNA-binding protein ChIP-seq



b Histone modification ChIP-seq



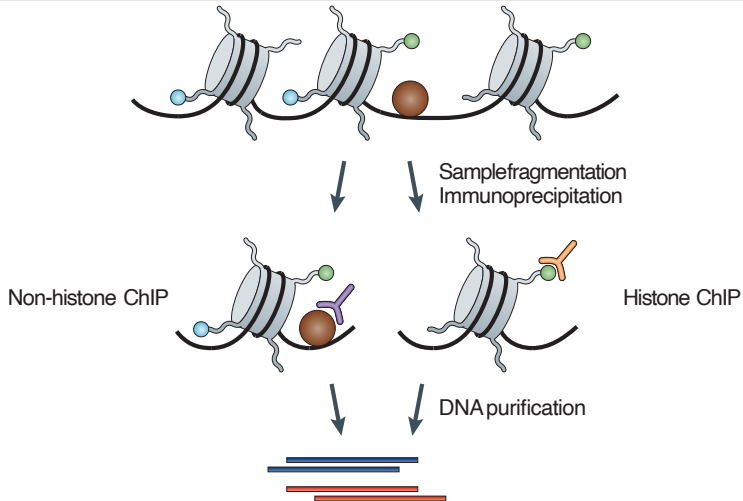
c DNase-seq



- Binding
- Sequencing

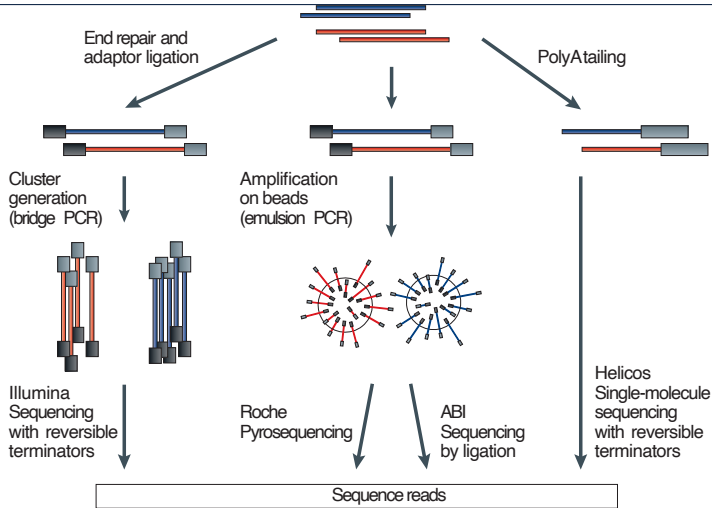
Furey (2012)

How ChIP-seq works



(Park 2009)

How ChIP-seq works



(Park 2009)

Peak calling: signal vs. background

Key idea: Find genomic regions with more reads than expected by chance

- Divide genome into bins (e.g., 200 bp)
- Count reads in each bin
- Compare ChIP vs Input (control)
- Statistical test: Is enrichment significant?

Common approaches:

- **MACS2:** Poisson model, local background
- **normR:** Binomial mixture model
- **DiffBind:** Differential binding analysis

Multiple testing: FDR correction across all bins!

ChIP-seq Peak Calling Exercise

```
library(AnnotationHub)
library(rtracklayer)

data_dir <- "../data/chipseq/"
dir.create(data_dir, showWarnings = FALSE, recursive = TRUE)

chr <- "chr21" # Focus on chr21 to keep data manageable

# Download ENCODE CTCF ChIP peaks
peaks_file <- paste0(data_dir, "ctcf_gm12878_peaks.rds")
if.nEEDED(peaks_file, {
  cat("Downloading ENCODE CTCF peaks...\n")
  ah <- AnnotationHub()
  query_peaks <- query(ah, c("CTCF", "GM12878", "narrowPeak", "ENCODE"))
  chip_peaks_all <- query_peaks[[1]]
  chip_peaks <- chip_peaks_all[seqnames(chip_peaks_all) == chr]
  saveRDS(chip_peaks, file = peaks_file)
})
chip_peaks <- readRDS(peaks_file)
```

ChIP-seq Peak Calling Exercise

```
# Download ENCODE DNase-seq peaks (open chromatin for comparison)
dnase_file <- paste0(data_dir, "dnase_gm12878_peaks.rds")
if (!file.exists(dnase_file)) {
  cat("Downloading ENCODE DNase-seq data...\n")
  ah <- AnnotationHub()
  query_dnase <- query(ah, c("DNase", "GM12878", "narrowPeak", "ENCODE"))
  if (length(query_dnase) > 0) {
    dnase_peaks_all <- query_dnase[[1]]
    dnase_peaks <- dnase_peaks_all[seqnames(dnase_peaks_all) == chr]
    saveRDS(dnase_peaks, file = dnase_file)
    cat(sprintf("Saved %d DNase peaks\n", length(dnase_peaks)))
  } else {
    cat("No DNase data found\n")
    saveRDS(GRanges(), file = dnase_file)
  }
}
dnase_peaks <- readRDS(dnase_file)
```

Peak Calling with normr

```
## Create genomic bins
binsize <- 1000
bins <- GRanges(chr, IRanges(start = seq(1, 48e6, binsize),
                               width = binsize))

## Count reads in bins
chip_counts <- countOverlaps(bins, chip_reads)
dnase_counts <- countOverlaps(bins, dnase_reads)

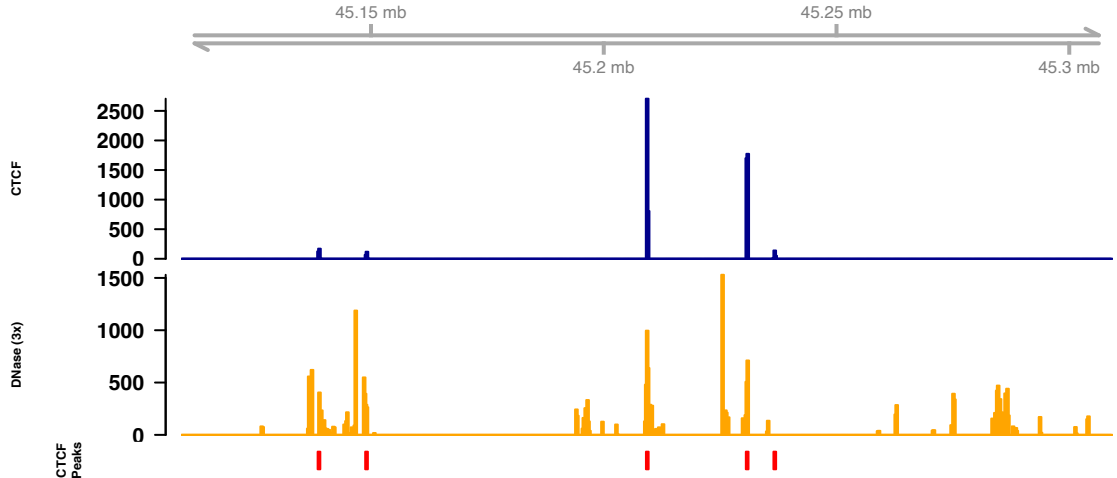
## Call peaks using normR
fit <- normr::enrichR(treatment = chip_counts,
                      control = dnase_counts,
                      genome = bins,
                      verbose = FALSE)

## Get significant peaks (FDR < 0.05)
peaks_called <- normr::getRanges(fit)
peaks_sig <- peaks_called[peaks_called$qvalue < 0.05]
```

- Foreground (CTCF) vs. background (control, DNase)
- Enrichment-based Binomial Tests
- Input: piled signals
- Output: genomic regions (BED)
- Narrow vs. broad peaks

Visualizing CTCF Binding Sites (using Gviz)

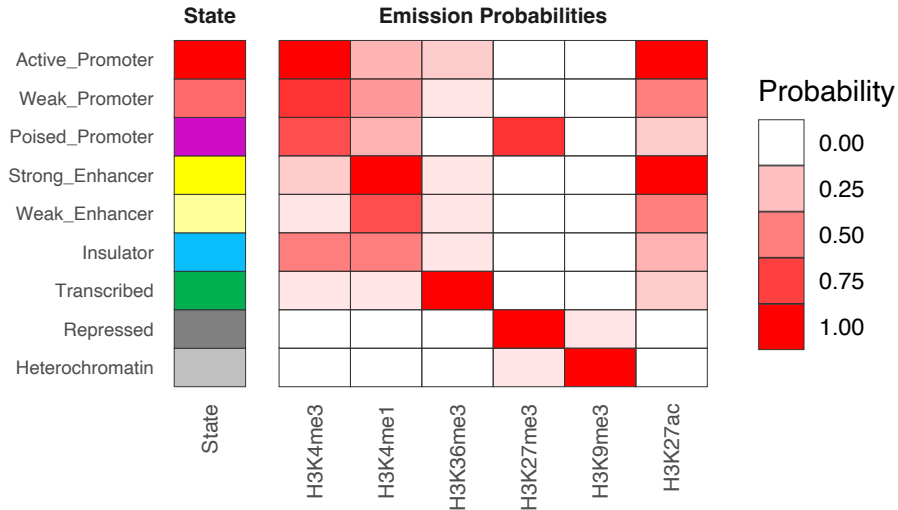
CTCF ChIP-seq on chr21:45,109,226-45,309,226



Key Concepts from ChIP-seq Peak Calling

- **Biological:** ChIP-seq identifies protein-DNA binding sites
- **Statistical** models:
 - Poisson/Binomial models for count data
 - Local background estimation
 - Multiple testing correction (FDR)
- **Computational** pipeline:
 - Millions of bins → massive multiple testing problem
 - Need efficient algorithms (MACS2, normr)
- **Quality control:**
 - Peak width distribution (sharp vs broad)
 - Fraction of reads in peaks (FRiP score)
 - Replicate concordance

ChromHMM: Emission Probabilities



ChromHMM: Integrating Multiple Histone Marks

How it works:

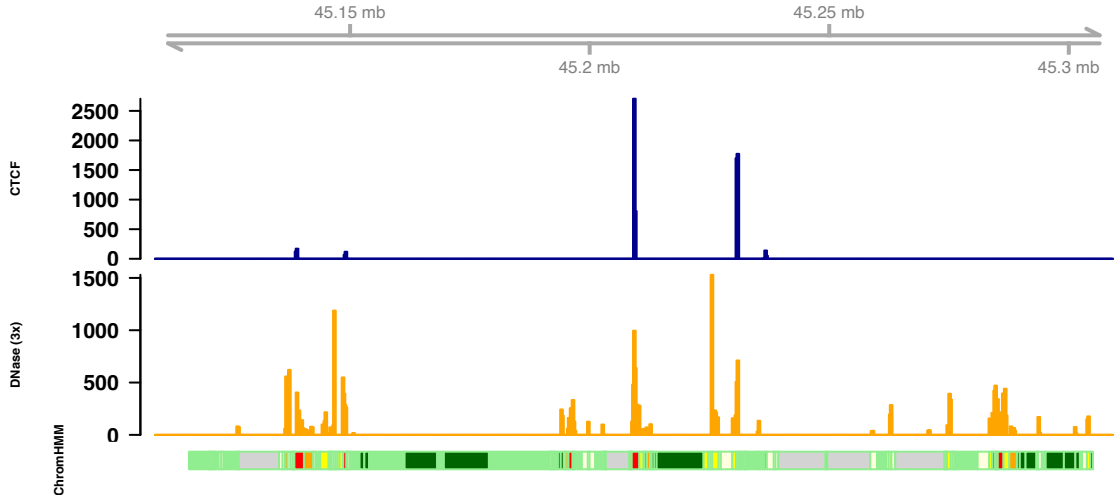
- Input: Multiple histone modification ChIP-seq signals
- Model: Hidden Markov Model learns chromatin states
- Each state defined by emission probabilities (previous slide)
- Output: Genome segmented into functional states

In our genome browser:

- ChromHMM track shows where each state occurs
- CTCF peaks often coincide with specific states (insulators, enhancers)

Visualizing CTCF Binding Sites (using Gviz)

CTCF ChIP-seq on chr21:45,109,226-45,309,226



Assay for Transposase-Accessible Chromatin (ATAC)

Goal: Map open chromatin regions genome-wide

Key insight: No antibody needed!

Protocol:

- 1 Tn5 transposase + DNA
- 2 Cuts accessible chromatin preferentially
- 3 Adds sequencing adapters
- 4 Sequence DNA fragments
- 5 Map reads to genome
- 6 **Call peaks**

Advantages:

- Fast protocol (~3 hours)
- Small cell numbers (500-50K cells)
- No antibodies required
- Genome-wide view

ATAC-seq vs. ChIP-seq

ATAC-seq:

- Measures **chromatin accessibility**
- All open regions (enhancers, promoters, etc.)
- No protein specificity
- No antibody needed
- Cheaper, faster

Use cases:

- Map all regulatory elements
- Compare cell states
- Find active enhancers
- TF footprinting

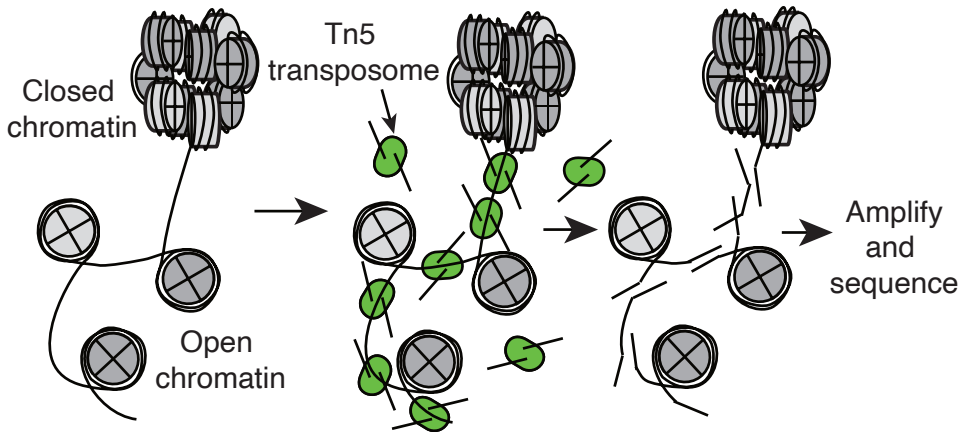
ChIP-seq:

- Measures **protein-DNA binding**
- Specific protein of interest
- Requires good antibody
- More expensive

Use cases:

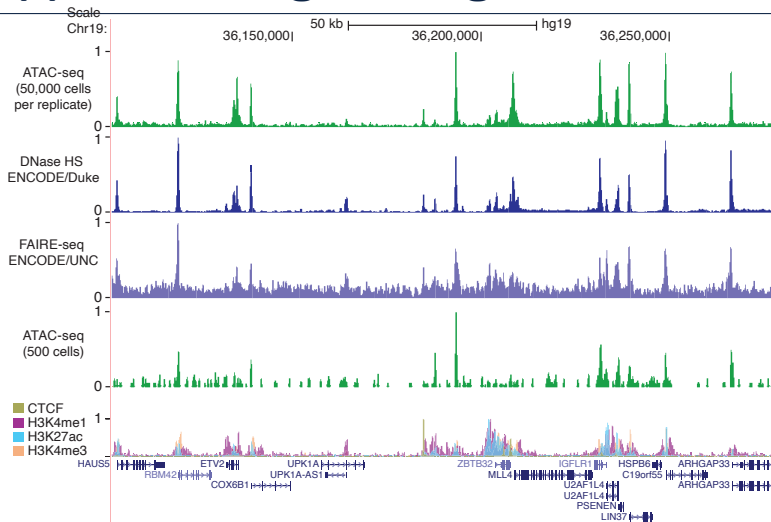
- Find TF binding sites
- Map histone modifications
- Validate ATAC peaks
- Mechanistic studies

ATAC-seq produces regular fragment sizes



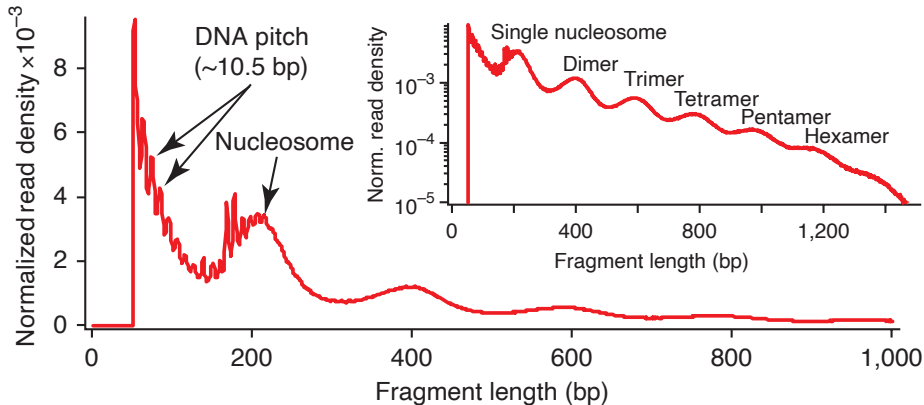
Tn5: cuts open DNA and insert adapter sequence.

ATAC-seq produces regular fragment sizes



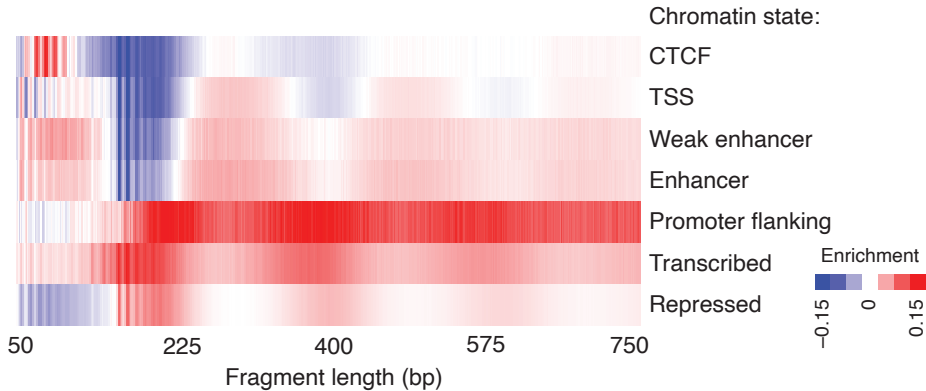
(Buenrostro et al. 2013)

ATAC-seq produces regular fragment sizes



(Buenrostro et al. 2013)

ATAC-seq produces regular fragment sizes



ATAC-seq enriches known gene regulatory elements in genome

(Buenrostro et al. 2013)

Differential Accessibility Analysis

```
# After peak calling, create count matrix
# Rows = genomic bins, Columns = samples
count_matrix <- matrix(...) # Read counts per bin per sample

colData <- data.frame( # Set up experimental design
  condition = factor(c(rep("Control", 3), rep("Treatment", 3)))
)

dds <- DESeqDataSetFromMatrix(countData = count_matrix, # Run differential accessibility
  colData = colData, # (same as RNA-seq!)
  design = ~ condition)

dds <- DESeq(dds)
results <- results(dds, contrast = c("condition", "Treatment", "Control"))

sig_regions <- results[results$padj < 0.05, ] # Get significant regions (FDR < 0.05)
```

Key point: Same statistical framework as RNA-seq (DESeq2)!

ATAC-seq Applications

1. Identify regulatory elements

- Promoters (at TSS)
- Enhancers (distal peaks)
- Silencers
- Insulators

2. Compare cell types/states

- Development
- Differentiation
- Disease vs healthy

3. TF footprinting

- Bound TF → protected DNA
- Dip in signal at TF motif
- Infer active TFs

4. Integration

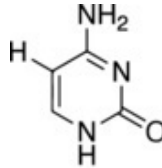
- ATAC + RNA-seq → **link enhancers to genes**
- ATAC + ChIP-seq → TF binding
- ATAC + GWAS → **functional variants**

Today's lecture

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- 2 DNA Methylation (credit: Keegan Korthauer)
- 3 Multiomic Data Integration with Regulatory Genomics

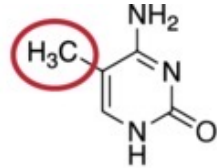
What is DNA Methylation?

- **5-methylcytosine (5mC)**: a methyl group added to cytosine at CpG dinucleotides
- **Heritable**: copied faithfully during DNA replication (by DNMT1)
- **Regulatory role**:
 - CpG islands at promoters → gene silencing when methylated
 - Enhancer activity correlates with hypomethylation
 - Imprinting, X-inactivation

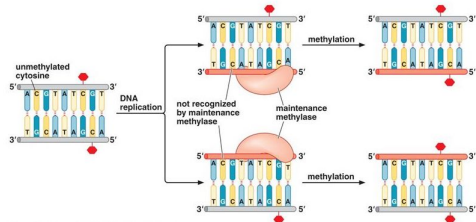


Cytosine

epigentek.com

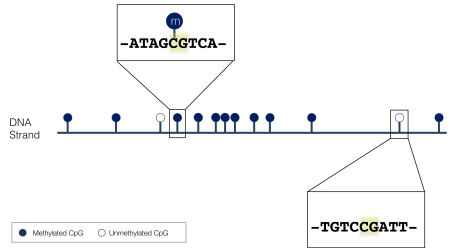
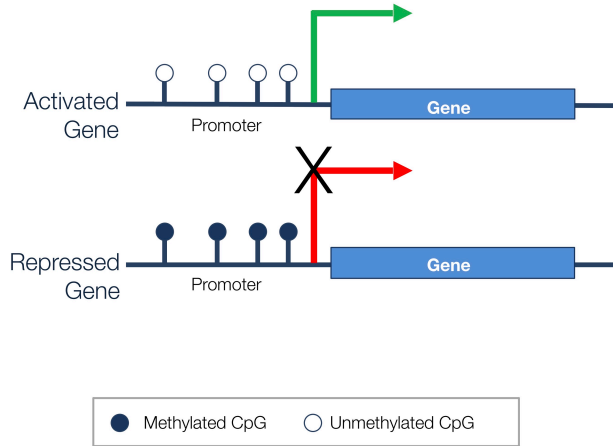


methylated Cytosine

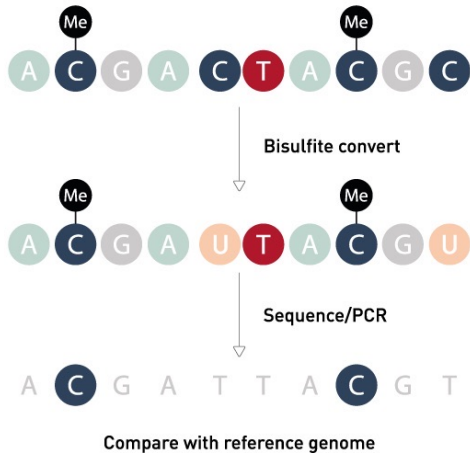


slideplayer.com

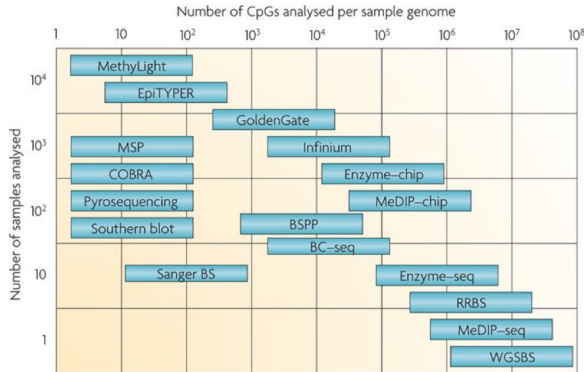
DNA Methylation and Gene Regulation



Measuring DNA Methylation

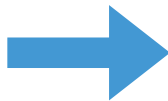
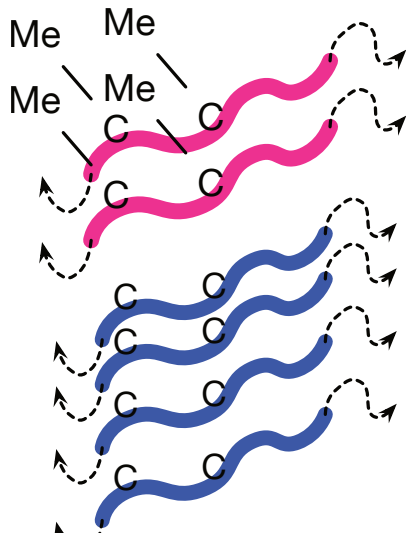


Bisulfite conversion (*diagenode.com*)

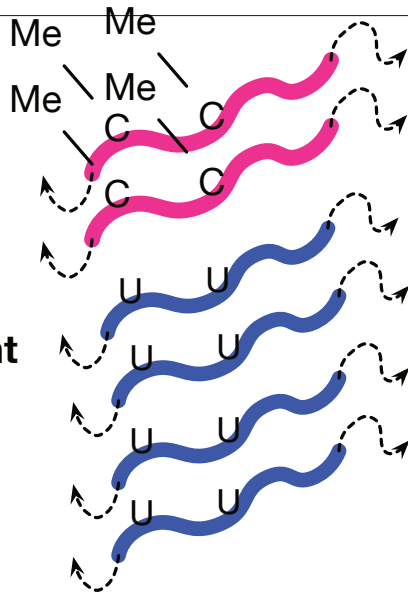


(Laird 2010)

Bisulfite Sequencing



**bisulfite
treatment**



Beta (β) values

- Combine two measurements for each CpG probe, measure level of methylation:

$$\beta = \frac{\text{methylated intensity}}{(\text{methylated intensity} + \text{unmethylated intensity})}$$

- Range from 0 to 1
- Easily interpretable as a **methylation proportion**
- Best used for **visualization**

M-values

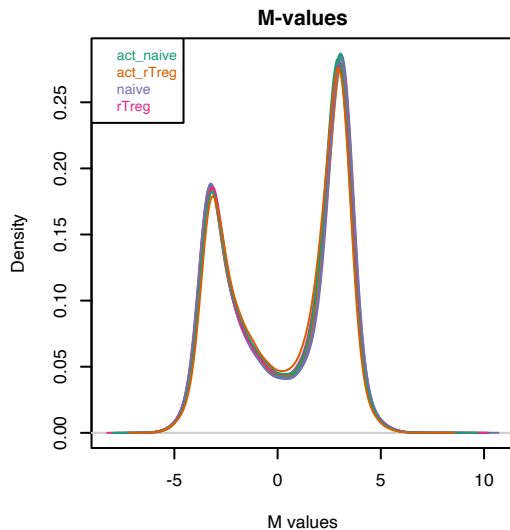
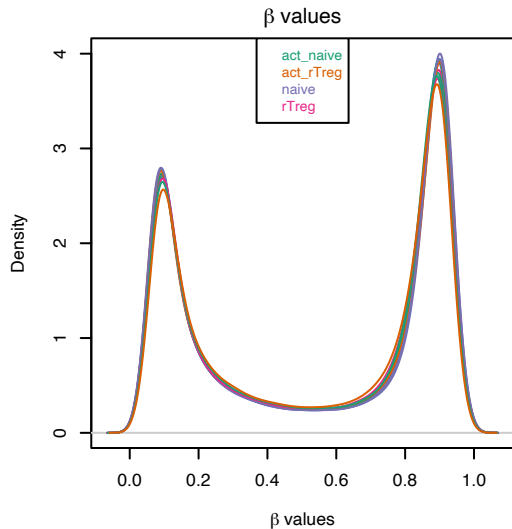
- Transformed β values:

$$M = \log_2 \left(\frac{\text{methylated intensity}}{\text{unmethylated intensity}} \right) = \log_2 \left(\frac{\beta}{1 - \beta} \right)$$

- The *logit* transformation — maps a proportion onto the real line
- Alleviates heteroscedasticity of β values (recall: Binomial(n, p) variance = $np(1 - p)$)
- Best used for **statistical testing**

(Du et al. 2010)

Comparing β and M values



Methylation Array Analysis Workflow

- Analysis strategy: **linear regression on M-values** (limma)
- Key R packages:
 - `minfi` — preprocessing and normalization of array data
 - `limma` — differentially methylated cytosines (DMC)
 - `dmrseq` / `bumphunter` — differentially methylated **regions** (DMR)
- Example dataset: sorted T cells (Zhang et al. 2013)
 - Naive vs. activated naive T cells
 - GEO: GSE49667

Differentially Methylated Cytosines (DMC)

```
library(limma)
cellType <- factor(targets$Sample_Group)
individual <- factor(targets$Sample_Source)
design <- model.matrix(~ cellType + individual, data = targets)
```

```
## Fit one linear model per CpG probe
```

```
mfit <- lmFit(mVals, design)
```

```
cfit <- eBayes(mfit)
```

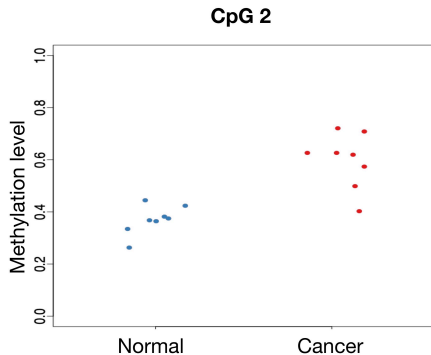
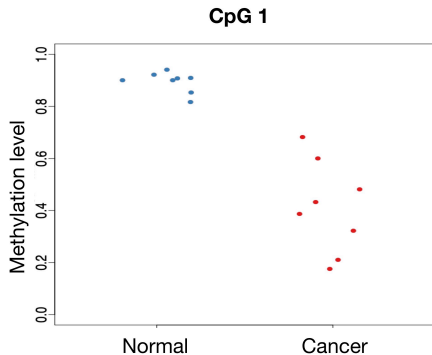
```
## Top differentially methylated CpGs (naive vs. act_naive)
```

```
topTable(cfit, coef = "cellTypenaive", number = 5)
```

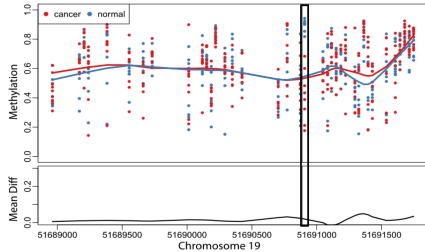
##		logFC	AveExpr	t	P.Value	adj.P.Val	B
##	cg15459165	-3.892074	-1.47895212	-22.09541	1.993958e-08	0.002793543	6.810420
##	cg19704755	-4.139885	0.09100138	-20.35779	3.787335e-08	0.002793543	6.594795
##	cg13531460	3.484448	-0.63132888	20.23020	3.978221e-08	0.002793543	6.577256
##	cg17137500	-3.007052	-2.47635924	-20.21569	4.000601e-08	0.002793543	6.575245
##	cg17048073	-3.650550	-0.87247048	-20.04737	4.270889e-08	0.002793543	6.551674

DMCs vs DMRs

- Individual CpG tests → **differentially methylated cytosines (DMCs)**
- Methylation of nearby CpGs is **correlated** and regulated in tandem
- Grouping nearby significant CpGs → **differentially methylated regions (DMRs)**

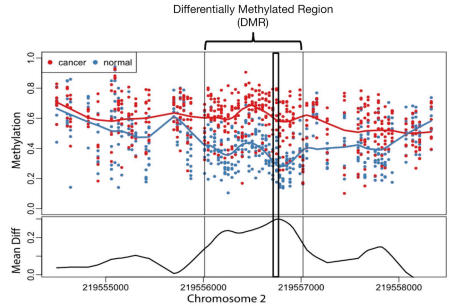


DMCs vs DMRs



Adapted from Jaffe et al., 2012 (*Int J Epidemiol*)

CpG 1



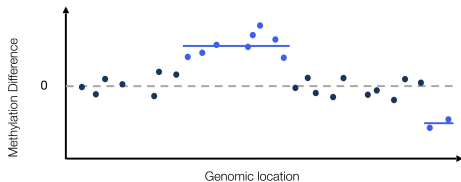
Adapted from Jaffe et al., 2012 (*Int J Epidemiol*)

CpG 2

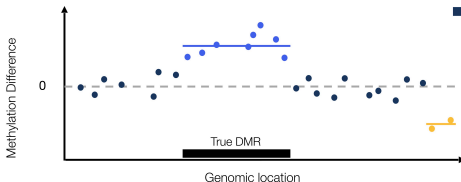
FDR at the Region Level

Results from testing individual CpGs

- Significant
- Not Significant



- True Positive (TP)
- False Positive (FP)
- True Negative (TN)



- Grouping significant CpGs post-hoc **inflates FDR**

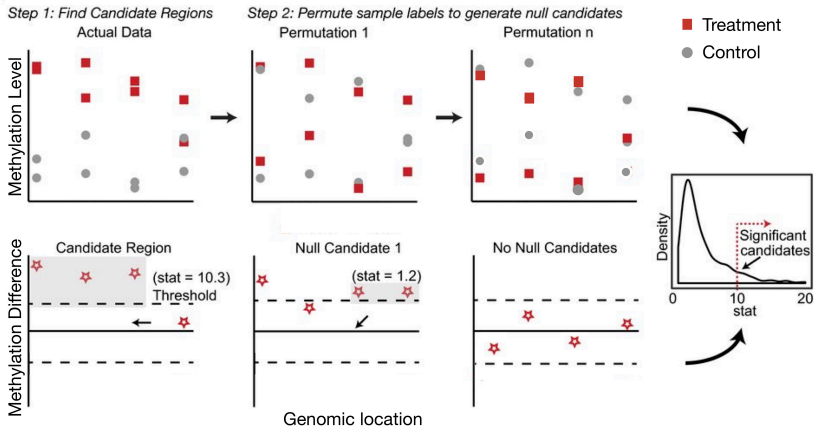
$$FDR_{CpG} = \frac{2}{10} = 0.20$$

$$FDR_{DMR} = \frac{1}{2} = 0.50$$

- Need to model region-level signal directly

BSmooth (Hansen et al. 2012), DSS (Park and Wu 2016)

Accurate DMR Inference: dmrseq

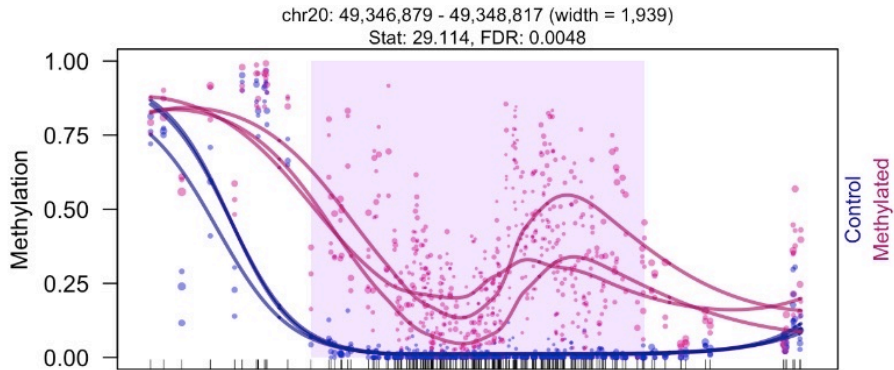


- Model correlated methylation signal over region
- Permutation for accurate FDR control



(Korthauer et al. 2019)

dmrseq Output



- *De novo* regions with FDR estimate
- Ranked by effect size of methylation difference
- Can adjust for covariates

Summary: DNA Methylation Analysis

- **Data:** Illumina arrays $\rightarrow$ per-CpG β values (proportions)
- **Model:** linear regression on logit-transformed M-values
 - Same `limma` framework as gene expression microarrays
 - Additive models, interactions, paired designs all apply
- **Output:**
 - **DMC** analysis: per-probe test (`limma`)
 - **DMR** analysis: region-level modelling (`dmrseq/bumphunter`)
- **Back-transform** estimates to β scale for biological interpretation:

$$\hat{\beta} = \frac{2^{\hat{M}}}{1 + 2^{\hat{M}}}$$

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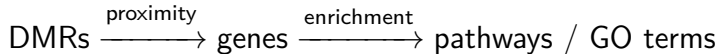
From DMRs to Biological Meaning

We found differentially methylated regions — now what?

Key questions:

- Which **genes** are near these regions?
- Which **biological processes** are affected?

Workflow:



This generalises to **any** set of genomic regions: ATAC peaks, ChIP peaks, GWAS loci

Standard Enrichment: A Problem

Naive approach: map each region to its nearest gene, collect gene list, run hypergeometric test

Why this fails:

- Gene-dense regions contribute many genes → **inflated hits**
- Distal enhancers (>5 kb) are silently ignored
- Background is “all genes”, not “all genomic space”

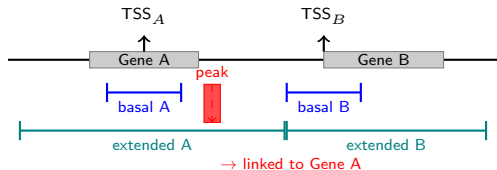
Example: a peak between a 50-gene cluster and an intergenic desert → the cluster dominates

Regions → Enrichment Directly

GREAT (McLean et al. 2010). Each gene gets a **regulatory domain**:

Basal domain: ± 5 kb around TSS (always kept)

Extended domain: expands up to 1 Mb, until the next gene's basal boundary



A region can be linked to **multiple** genes; background = all genomic space

GREAT: Why It Works Better

	Standard	GREAT
Input	gene list	genomic regions
Background	all genes	all genome
Long-range	ignored	up to 1 Mb
Gene density bias	yes	corrected

Useful for: ATAC peaks, ChIP peaks, DMRs, GWAS loci

GREAT Step 1: Prepare Input Regions

```
library(rGREAT)
library(TxDb.Hsapiens.UCSC.hg19.knownGene)

## chip_peaks loaded earlier - keep standard chromosomes
std_chrs <- paste0("chr", c(1:22, "X", "Y"))
ctcf_gr <- GRanges(as.character(seqnames(chip_peaks)),
                  IRanges(start(chip_peaks), end(chip_peaks)))
ctcf_gr <- ctcf_gr[as.character(seqnames(ctcf_gr)) %in% std_chrs]
cat(length(ctcf_gr), "CTCF peaks ready\n")
```

446 CTCF peaks ready

- Regions = CTCF ChIP-seq peaks (GM12878, whole genome)
- Same syntax for DMRs, ATAC peaks, or GWAS loci

GREAT Step 2: Run Enrichment

```
great_file <- "../data/great_ctcf_gobp.rds"
if.needed(great_file, {
  great_res <- great(ctcf_gr,
                    gene_sets = "GO:BP",
                    tss_source = "TxDb.Hsapiens.UCSC.hg19.knownGene",
                    verbose    = FALSE)
  saveRDS(great_res, great_file)
})
great_res <- readRDS(great_file)
```

- gene_sets: "GO:BP", "GO:MF", "KEGG", "MSigDB:H", ...
- tss_source: genome-specific gene model
- Returns a GreatObject (region-gene links + enrichment)

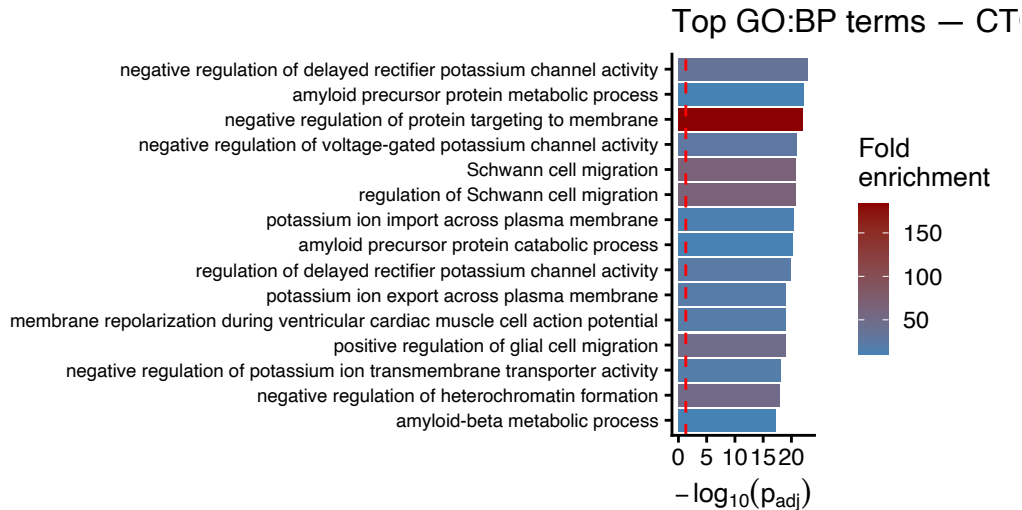
GREAT Step 3: Inspect Results

```
tb <- getEnrichmentTable(great_res)
cols <- c("description", "observed_region_hits",
          "fold_enrichment", "p_adjust")
head(tb[order(tb$p_adjust), cols], 6)
```

```
##                                     description
## 1 negative regulation of delayed rectifier potassium channel activity
## 2                               amyloid precursor protein metabolic process
## 3               negative regulation of protein targeting to membrane
## 4   negative regulation of voltage-gated potassium channel activity
## 5                               Schwann cell migration
## 6               regulation of Schwann cell migration
##   observed_region_hits fold_enrichment   p_adjust
## 1                   22      36.12004 1.143964e-23
## 2                   38      10.23917 6.198902e-23
## 3                   13     183.74047 9.691236e-23
## 4                   22      27.65168 8.614990e-22
## 5                   16      66.46148 1.283780e-21
## 6                   16      66.46148 1.283780e-21
```

- fold_enrichment: observed / expected overlap, p_adjust: BH-corrected p-value

GREAT Step 4: Visualize Top Terms



GWAS Variants Are Mostly Non-Coding

Disease GWAS identifies thousands of associated variants

The puzzle: >90% of hits fall outside protein-coding exons

Key questions:

- Where do these variants **act**?
- Which **cell types** are disease-relevant?

Strategy: overlap GWAS loci with cell-type-specific epigenetic annotations

GWAS → Epigenetic States: The Approach

- ① Collect GWAS summary statistics (SNPs with $P < 5 \times 10^{-8}$)
- ② Download cell-type-specific chromatin annotations (e.g., Roadmap Epigenomics: 127 cell/tissue types)
- ③ For each cell type: test whether GWAS SNPs are **enriched** in active enhancers
- ④ The most enriched cell type = likely disease-relevant tissue

Statistical test: Fisher's exact test (or LD-score regression)

GWAS Step 1: Fetch RA Loci from GWAS Catalog

```
library(gwasrapidd)
ra_file <- "../data/ra_gwas_hits.rds"
if (!file.exists(ra_file)) {
  assoc <- get_associations(study_id = "GCST90132222")
  variants <- get_variants(study_id = "GCST90132222")
  ra <- merge(assoc@risk_alleles[, c("association_id", "variant_id")],
             variants@variants[, c("variant_id", "chromosome_name",
                                   "chromosome_position")],
             by = "variant_id")
  ra$CHR <- paste0("chr", ra$chromosome_name)
  saveRDS(ra, ra_file)
}
ra <- readRDS(ra_file)
cat(nrow(ra), "genome-wide significant RA loci (GCST90132222)\n")
```

```
## 144 genome-wide significant RA loci (GCST90132222)
```

- Study: 22,350 European ancestry samples (Okada et al. 2014)
- Coordinates: hg19 / GRCh37

GWAS Step 2: Build Locus Windows

```
## Expand each index SNP  $\pm 50$  kb (covers typical LD block)
sig_gr <- GRanges(ra$CHR,
                  IRanges(ra$chromosome_position - 50000,
                          ra$chromosome_position + 50000))
sig_gr <- sig_gr[start(sig_gr) > 0]

## Chromosome-proportional random background (hg19)
set.seed(42)
chr_sizes <- seqlengths(BSgenome.Hsapiens.UCSC.hg19::
                        BSgenome.Hsapiens.UCSC.hg19)[paste0("chr", 1:22)]
chr_s <- sample(names(chr_sizes), 5e4, replace = TRUE,
               prob = chr_sizes / sum(chr_sizes))
pos_s <- mapply(function(ch) sample(chr_sizes[ch] - 1e5, 1) + 50000, chr_s)
bg_gr <- GRanges(chr_s, IRanges(pos_s - 50000, pos_s + 50000))
cat(length(sig_gr), "locus windows,", length(bg_gr), "background windows\n")

## 144 locus windows, 50000 background windows
```

GWAS Step 3: Load Roadmap H3K27ac Enhancers

```
enh_dir    <- "../data/roadmap/"
enh_labels <- c(CD4_Naive  = "AH43615", CD4_Memory = "AH43684",
               CD19_Bcell = "AH44900", CD14_Mono  = "AH44890",
               Adult_Liver= "AH43458", Brain     = "AH43501")

## Load from locally cached RDS files (downloaded once via AnnotationHub)
enhancers <- lapply(names(enh_labels), function(ct)
  readRDS(paste0(enh_dir, ct, "_H3K27ac.rds")))
names(enhancers) <- names(enh_labels)
immune_cells <- c("CD4_Naive", "CD4_Memory", "CD19_Bcell", "CD14_Mono")
sapply(enhancers, length) # peaks per cell type
```

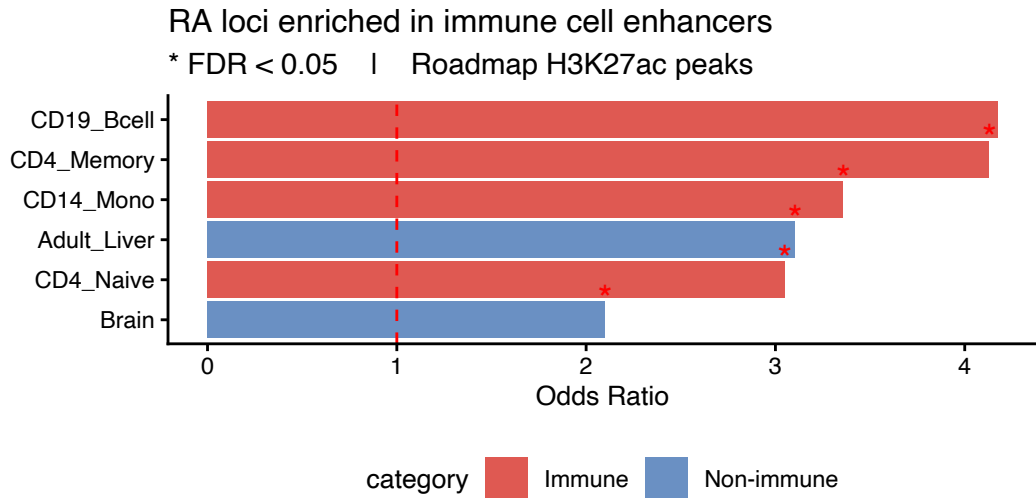
##	CD4_Naive	CD4_Memory	CD19_Bcell	CD14_Mono	Adult_Liver	Brain
##	66232	92684	65749	61058	121020	188662

- H3K27ac = active enhancer / promoter mark
- Roadmap Epigenomics: 127 reference human epigenomes (Kundaje et al. 2015)

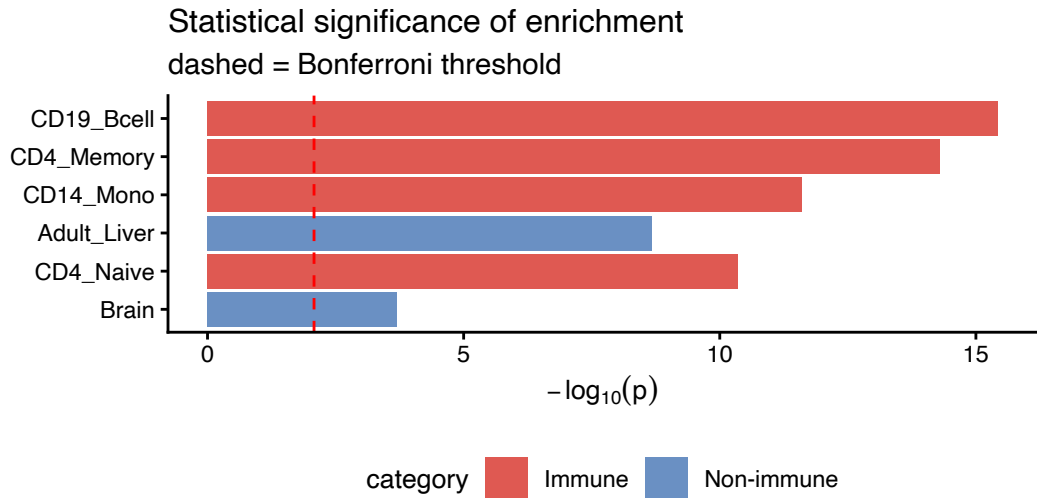
GWAS Step 4: Fisher's Exact Test per Cell Type

```
enr_file <- "../data/gwas_ra_enrichment.rds"
if.needed(enr_file, {
  enr <- do.call(rbind, lapply(names(enhancers), function(ct) {
    enh <- enhancers[[ct]]
    a <- sum(countOverlaps(sig_gr, enh) > 0)
    b <- length(sig_gr) - a
    c <- sum(countOverlaps(bg_gr, enh) > 0)
    d <- length(bg_gr) - c
    ft <- fisher.test(matrix(c(a, b, c, d), 2))
    data.frame(cell_type = ct, OR = ft$estimate,
               pval = ft$p.value, frac = a / length(sig_gr))
  )))
  enr$fdr <- p.adjust(enr$pval, "BH")
  enr$category <- ifelse(enr$cell_type %in% immune_cells, "Immune", "Non-immune")
  saveRDS(enr, enr_file)
})
enr <- readRDS(enr_file)
enr$cell_type <- factor(enr$cell_type, levels = enr$cell_type[order(enr$OR)])
```

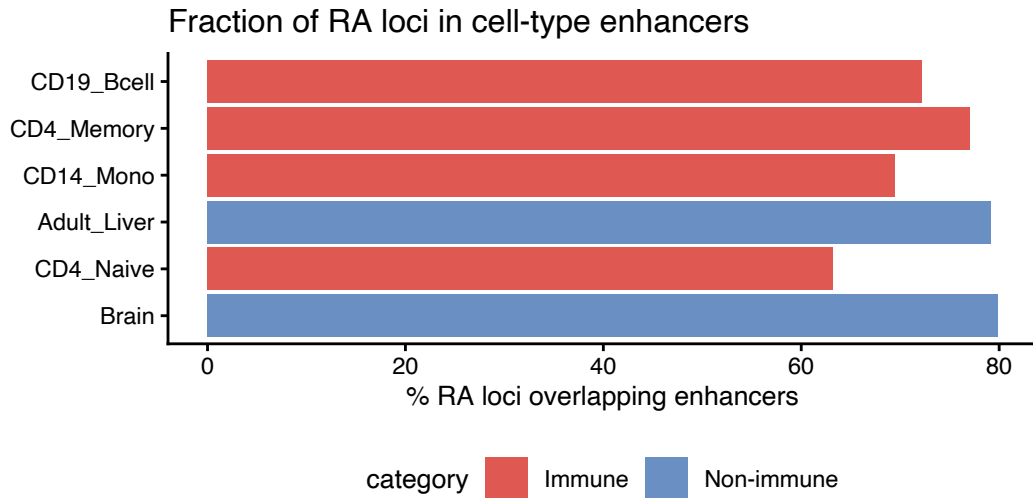
GWAS Enrichment: Odds Ratio



GWAS Enrichment: Statistical Significance



GWAS Enrichment: Overlap Fraction



Limitation: Fisher's Test Ignores LD

The problem: nearby SNPs are correlated through **linkage disequilibrium (LD)**

Two significant SNPs 20 kb apart may represent **one causal signal**, not two

⇒ overlapping the same enhancer window gets double-counted

Consequence: Fisher's test **inflates enrichment** in gene-dense, LD-rich regions

Better approaches:

- **Stratified LD score regression (S-LDSC)** — partitions heritability by annotation, models LD (Finucane et al. 2015)
- **GARFIELD** — permutation matching on MAF, distance to TSS, LD
- **GoShifter** — circular permutation within LD blocks

Real-World GWAS–Epigenome Findings (examples)

Rheumatoid arthritis → CD4⁺ T cells

adaptive immunity

IBD → gut immune cells

mucosal inflammation

Type 2 diabetes → pancreatic islets

β-cell dysfunction

Schizophrenia → neurons

brain circuit defects

Height → chondrocytes

cartilage growth plates

(Kundaje et al. 2015)

Why GWAS–Epigenome Integration Matters

Identify disease-relevant cell types

- Guides which cells to study experimentally
- 90% of hits are non-coding — context is everything

Mechanistic insight

- Variant disrupts an enhancer → changes TF binding → alters gene expression

Therapeutic direction

- Target the right cell type, not just the right gene

Links to scRNA-seq

- Fine-map to specific cell states within a tissue

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